

Spectral-rich sharpness for a second-order semigroup approximation estimate

FA/Banach run packet for arXiv:1801.06749

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Source problem

The source paper is A. Gomilko, S. Kosowicz, and Yu. Tomilov, *A general approach to approximation theory of operator semigroups*, arXiv:1801.06749, published in *Journal de Mathématiques Pures et Appliquées* 127 (2019), 216–267.

Corollary 4.10 of the source paper states that, if $-A$ generates a bounded C_0 -semigroup on X , $(g_t)_{t>0} \subset \mathcal{B}_4$, and $M = \sup_{t \geq 0} \|e^{-tA}\|$, then for all $t > 0$, $n \in \mathbb{N}$, and $x \in \text{dom}(A^3)$,

$$\|g_t^n(tA/n)x - e^{-tA}x - (2n)^{-1}(g_t''(0) - 1)t^2e^{-tA}A^2x\| \leq MC(g_t)t^3n^{-3/2}\|A^3x\|.$$

After proving sharpness only for a shift semigroup on $C_0(\mathbb{R})$ and Euler's formula, the authors state that the optimality of this estimate should hold in a more general context.

Replacing $b[g_n]$ and $d_0[g_n]$ by their expressions in terms of the derivatives of g at zero, we formulate below Theorem [4.9](#) in a more explicit form.

Corollary 4.10. *Let $-A$ be the generator of a bounded C_0 -semigroup $(e^{-tA})_{t \geq 0}$ on X , and let $(g_t)_{t>0} \subset \mathcal{B}_4$. Let $M := \sup_{t \geq 0} \|e^{-tA}\|$. Then for all $t > 0$, $n \in \mathbb{N}$, and $x \in \text{dom}(A^3)$,*

$$(4.30) \quad \|g_t^n(tA/n)x - e^{-tA}x - (2n)^{-1}(g_t''(0) - 1)t^2e^{-tA}A^2x\| \leq MC(g_t)t^3n^{-3/2}\|A^3x\|,$$

where

$$C(g_t) = \left(\frac{(g_t''(0) - 1)(g_t'''(0) - 1)}{2} \right)^{1/2}.$$

Moreover, for all $t > 0$, $n \in \mathbb{N}$, and $x \in \text{dom}(A^4)$,

$$(4.31) \quad \|g_t^n(tA/n)x - e^{-tA}x - (2n)^{-1}(g_t''(0) - 1)t^2e^{-tA}A^2x\| \leq MC_1(g_t)t^3n^{-2}(\|A^3x\| + t\|A^4x\|),$$

where

$$C_1(g_t) = g_t'''(0) - 1.$$

By (4.34), we have $\sqrt{n}\omega(1/\sqrt{n}, A^2 f) \leq 2$, so we can rewrite (4.39) in the form

$$(4.40) \quad \liminf_{n \rightarrow \infty} \frac{n}{\omega(1/\sqrt{n}, A^2 f)} \|R_{1,n} f\| \geq \frac{1}{6\sqrt{2\pi}}.$$

The inequality (4.40) shows that (4.33) is sharp. Consequently, (4.32) is sharp too.

We believe that the optimality of (4.30) can be shown in a more general context. However we feel that the exposition will then be overloaded by unnecessary technicalities, and postpone a general argument to another occasion.

This packet proves such a general-context sharpness result under the same spectral-rich hypothesis used in Proposition 4.11 of the source paper. The result is partial because it treats a fixed g and a natural spectral class, not all possible bounded semigroups or all varying families (g_t) .

Idea of the proof

The source proof of first-order optimality tests the HP functional calculus on large imaginary spectral points of size $\sqrt{n}/t$. The same test is exactly tuned to the second-order remainder. Along $z_n = \pm i\sqrt{n}$, Taylor expansion gives

$$n \log g(z_n/n) = -z_n - a[g] + O(n^{-1/2}),$$

because $z_n^2/n = -1$. After subtracting the second-order correction, the scalar remainder tends to $e^{-a[g]} - 1 + a[g]$, which is positive whenever $g \neq e^{-z}$. Spectral mapping then transfers this nonzero scalar boundary value into an operator-norm lower bound of order $t^3 n^{-3/2}$.

Notation

For $g \in \mathcal{B}_2$ put

$$a[g] = \frac{g''(0) - 1}{2}.$$

The source paper observes that $a[g] \geq 0$, and that $a[g] = 0$ if and only if $g(z) \equiv e^{-z}$. For fixed $g \in \mathcal{B}_4$ define the scalar second-order remainder

$$q_{t,n}(z) = g^n(tz/n) - e^{-tz} - a[g] n^{-1} t^2 z^2 e^{-tz}.$$

The regularized remainder multiplier is

$$h_{t,n}(z) = z^{-3} q_{t,n}(z),$$

with the removable value at $z = 0$. This is the HP-calculus multiplier corresponding to A^{-3} times the second-order remainder, understood in the same extended-calculus sense used in Proposition 4.11 of the source paper.

Main result

Theorem 1. *Let $-A$ be the generator of a bounded C_0 -semigroup on a Banach space X , and assume $\overline{\text{ran}(A)} = X$. Suppose that*

$$\{ |s| : s \in \mathbb{R}, is \in \sigma(A) \} = \mathbb{R}_+.$$

Let $g \in \mathcal{B}_4$ and $g(z) \not\equiv e^{-z}$. Then for every $t > 0$,

$$\liminf_{n \rightarrow \infty} n^{3/2} t^{-3} \|h_{t,n}(A)\| \geq e^{-a[g]} - 1 + a[g] > 0.$$

Consequently, in this spectral-rich class the $t^3 n^{-3/2}$ rate in Corollary 4.10 cannot be replaced by $o(t^3 n^{-3/2})$ for the regularized third-order norm of the second-order remainder.

Proof. Set $a = a[g]$. Since $g \neq e^{-z}$, the moment identity for functions in $\mathcal{B}_2$ gives $a > 0$. By the definition of $\mathcal{B}_4$, the boundary Taylor expansion at the origin is

$$g(w) = 1 - w + \frac{g''(0)}{2} w^2 + O(|w|^3).$$

For all sufficiently small w , $g(w)$ is nonzero and hence

$$\log g(w) = -w + aw^2 + O(|w|^3).$$

Fix $t > 0$. By the spectral hypothesis, for each n we may choose $\varepsilon_n \in \{-1, 1\}$ such that

$$\lambda_n = \varepsilon_n i \sqrt{n}/t \in \sigma(A).$$

Put $z_n = t\lambda_n = \varepsilon_n i \sqrt{n}$ and $w_n = z_n/n$. Then

$$n \log g(w_n) = -z_n + a \frac{z_n^2}{n} + O(n|w_n|^3) = -z_n - a + O(n^{-1/2}),$$

because $z_n^2/n = -1$. Hence

$$g^n(z_n/n) = e^{-z_n} (e^{-a} + O(n^{-1/2})).$$

Therefore

$$\begin{aligned} q_{t,n}(\lambda_n) &= g^n(z_n/n) - e^{-z_n} - a n^{-1} z_n^2 e^{-z_n} \\ &= e^{-z_n} (e^{-a} - 1 + a + O(n^{-1/2})). \end{aligned}$$

Since $|e^{-z_n}| = 1$, it follows that

$$|q_{t,n}(\lambda_n)| \longrightarrow e^{-a} - 1 + a.$$

The scalar function $h_{t,n}$ belongs to the HP algebra after removing the singularity at the origin: the numerator has vanishing Taylor coefficients through order two, and the integral representation behind Corollary 4.10 gives the corresponding bounded HP multiplier. Applying the spectral mapping theorem for the HP calculus, as in Proposition 4.11 of the source paper, gives

$$\|h_{t,n}(A)\| \geq r(h_{t,n}(A)) \geq |h_{t,n}(\lambda_n)|.$$

But $|\lambda_n|^{-3} = t^3 n^{-3/2}$, so

$$n^{3/2} t^{-3} |h_{t,n}(\lambda_n)| = |q_{t,n}(\lambda_n)| \longrightarrow e^{-a} - 1 + a.$$

The displayed lower bound follows. Finally $e^{-a} - 1 + a > 0$ for every $a > 0$, completing the proof. $\square$

Verifier report

The proof is analytic. As a sanity check, the script `code/scalar_asymptotic_check.py` evaluates the scalar residual for Euler’s formula $g(z) = (1 + z)^{-1}$, where $a[g] = 1/2$, along $z_n = i\sqrt{n}$. The output converges to $e^{-1/2} - 1 + 1/2$, as predicted by the proof.

Limitations and novelty status

This is not a full theorem for arbitrary bounded C_0 -semigroups. It proves optimality for the spectral-rich class $\{|s| : is \in \sigma(A)\} = \mathbb{R}_+$, the same kind of hypothesis used by the source paper for first-order optimality. Concrete examples are available, for instance multiplication generators with spectrum $i\mathbb{R}$ on $L^2(\mathbb{R})$.

Before promotion, the run indexes were searched for arXiv:1801.06749, the label **RowA0++**, and close phrases involving second-order semigroup optimality and $n^{-3/2}$. Web searches on 2026-07-18 for the exact title, the deferred-general-argument phrase, and close scalar-remainder phrases found the source paper and mirrors but no later exact answer. Novelty confidence is therefore moderate-high, subject to human review.

Human-review recommendation

Review the HP-calculus regularization line in the proof: the scalar asymptotic and spectral choice are straightforward, and the only technical point is that the regularized multiplier $h_{t,n}$ is legitimate in the same functional calculus setting as Proposition 4.11. If accepted, this packet supplies the general optimality argument that the source paper deferred for Corollary 4.10 within the natural spectral-rich class.

References

- [1] A. Gomilko, S. Kosowicz, and Yu. Tomilov, *A general approach to approximation theory of operator semigroups*, Journal de Mathématiques Pures et Appliquées 127 (2019), 216–267. arXiv:1801.06749.